12TH BIOLOGY · CHAPTER 23 · FULL CHAPTER NOTES

Pharmacology

Topic-wise short and long questions with answers from the complete chapter

INTRODUCTION

Q.1 [Short] Define pharmacology and drugs.

Pharmacology is the branch of biology that deals with the study of drugs and their effects on living organisms. It includes the sources, properties, composition, therapeutic uses and effects of drugs on the body. **Drugs** are chemical substances used to diagnose, prevent or treat diseases. They may be obtained from natural sources (plants, animals, microorganisms) or synthesized in laboratories.

Q.2 [Short] What is phytotherapy?

Medicines prepared from plants are commonly used in traditional systems of treatment. They are known as herbal medicines, and this treatment is called phytotherapy.

23.1 DRUG DISCOVERY AND DEVELOPMENT PROCESS

Q.3 [Short] Why are drug discovery and development important?

They help in the production of safe and effective medicines. The process begins with identifying disease-causing targets and developing compounds that interact with them. Before approval, a drug passes several stages of testing for safety, efficacy and quality. Scientists, researchers and regulatory authorities work together throughout.

Q.4 [Long] Describe the stages of drug discovery.

Drug discovery is the initial stage of finding a new drug for a disease. Its stages are:

1. **Target identification:** Scientists identify a specific molecule, such as a protein or enzyme, that is involved in causing the disease and can be affected by a drug.
2. **Target validation:** Researchers confirm by experiments that the target plays an important role in the disease and that acting on it can produce therapeutic effects.
3. **Lead compound identification:** Thousands of natural or synthetic compounds are screened to find substances that interact effectively with the target. These promising substances are called **lead compounds**.
4. **Lead optimization:** The lead compounds are modified to improve their effectiveness, reduce toxicity and enhance their drug-like properties.

Q.5 [Short] What are lead compounds? What is lead optimization?

Lead compounds are the promising substances, found by screening thousands of natural or synthetic compounds, that interact effectively with the target molecule. In lead optimization they are modified to improve effectiveness, reduce toxicity and enhance drug-like properties.

Q.6 [Short] Differentiate between target identification and target validation.

In **target identification**, a specific molecule (protein or enzyme) involved in causing the disease is identified. In **target validation**, experiments confirm that this target really plays an important role in the disease and that acting on it gives a therapeutic effect.

Q.7 [Long] Describe the stages of the drug developmental process.

Drug development is the process of testing and preparing the discovered drug for public use. Its stages are:

1. **Preclinical research:** The compounds are tested *in vitro* (cell-based culture) and *in vivo* (animals) to measure safety, toxicity and efficacy before human trials.
2. **Clinical trials:** The drug is tested on humans in three phases (see the table).
3. **FDA drug review:** Regulatory agencies (FDA or EMA) examine all the data of the clinical trials and approve or reject the drug before it is supplied to the market.
4. **Manufacturing and marketing:** The approved drug is produced on a large scale in pharmaceutical industries and made available for medical use.
5. **Post-marketing surveillance:** After release, the drug is continuously monitored for long-term effects and rare side effects.

Phase	Who is tested	Purpose
Phase 1	Small group (20–100) of healthy volunteers	To determine safety and dosage
Phase 2	Larger group (100–300) of patients	To test effectiveness and side effects
Phase 3	Large-scale trials (1,000–3,000)	To confirm efficacy and monitor adverse reactions

Q.8 [Short] What is preclinical research? Differentiate between in vitro and in vivo testing.

Preclinical research is the testing of compounds for safety, toxicity and efficacy before human trials. **In vitro** testing is done in cell-based cultures; **in vivo** testing is done in animals.

Q.9 [Short] Describe the three phases of clinical trials.

Phase	Who is tested	Purpose
Phase 1	Small group (20–100) of healthy volunteers	To determine safety and dosage
Phase 2	Larger group (100–300) of patients	To test effectiveness and side effects
Phase 3	Large-scale trials (1,000–3,000)	To confirm efficacy and monitor adverse reactions

Q.10 [Short] What is post-marketing surveillance? Why is it needed?

Even after its release, the drug is continuously monitored for long-term effects and rare side effects. This ensures ongoing safety in the population, because such effects may not appear during the trials.

Q.11 [Short] What is the role of the FDA or EMA?

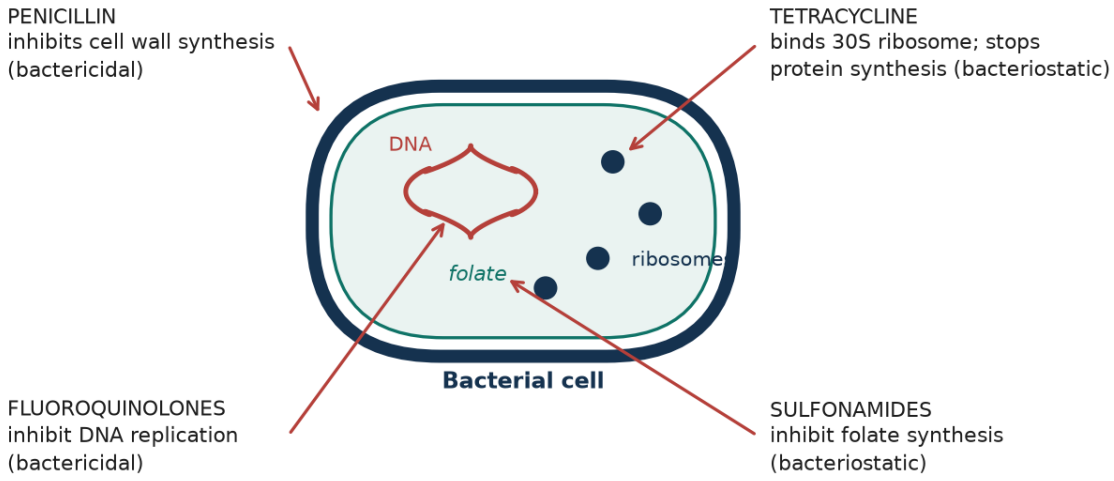
These regulatory agencies thoroughly examine all the data from clinical research trials. They approve or reject the drug before it is supplied to the market.

23.2 CLASSES OF ANTIBIOTIC

Q.12 [Short] What are antibiotics? Name their four major classes.

Antibiotics are drugs that treat bacterial infections by killing bacteria or stopping their growth. They are classified mainly by their mechanism of action and chemical structure. The four major classes are penicillin, tetracycline, fluoroquinolones and sulfonamides.

Class	Source / discovery	Effect	Target in bacteria
Penicillin	*Penicillium* fungi; Alexander Fleming (1928)	Bactericidal	Cell wall synthesis
Tetracycline	Soil bacteria of genus *Streptomyces*	Bacteriostatic	Protein synthesis (30S ribosomal subunit)
Fluoroquinolones	Synthetic; George Lesher (1962)	Bactericidal	DNA replication (type II topoisomerases)
Sulfonamides	Synthetic, from sulfanilamide; Gerhard Domagk (1932)	Bacteriostatic	Folic acid (folate) synthesis



Targets of the four classes of antibiotics in a bacterial cell

Q.13 [Short] Differentiate between bactericidal and bacteriostatic antibiotics.

Bactericidal	Bacteriostatic
Kill the bacteria.	Stop bacteria from growing and multiplying; the immune system destroys them.
Examples: penicillin, fluoroquinolones.	Examples: tetracycline, sulfonamides.

Q.14 [Long] Describe penicillin and its mode of action.

Penicillin is a group of bactericidal antibiotics derived from *Penicillium* fungi. It treats bacterial infections by inhibiting bacterial cell wall synthesis. Alexander Fleming discovered it in 1928. It is highly effective against gram-positive bacteria, including streptococcal and staphylococcal infections. Common side effects: nausea, vomiting, diarrhoea, abdominal pain and skin rashes.

Mode of action

1. **Targeting bacteria:** It targets bacteria only, because animal cells lack the cell wall that the drug attacks.
2. **Cell wall inhibition:** It interrupts the cross-linking of peptidoglycan (murein) chains, so new cell wall is not synthesised.
3. **Lysing the bacterial cell:** The cell wall can no longer maintain its structural integrity, and the cell bursts (lysis). So it kills bacteria rather than just inhibiting growth.
4. **Resistance mechanism:** Some bacteria defend themselves by producing specific enzymes such as beta-lactamases.

Q.15 [Short] Why does penicillin not harm human cells?

Penicillin attacks the bacterial cell wall by stopping the cross-linking of peptidoglycan chains. Animal (human) cells have no cell wall, so there is nothing in them for the drug to attack.

Q.16 [Short] How do some bacteria resist penicillin?

They defend themselves by producing specific enzymes such as beta-lactamases.

Q.17 [Long] Describe tetracycline and its mode of action.

Tetracycline is derived from soil-dwelling bacteria of the genus *Streptomyces*. It is a bacteriostatic oral antibiotic used for bacterial infections like chlamydia and syphilis. It inhibits bacterial protein synthesis. It should not be taken with dairy or iron supplements. Common side effects: nausea, vomiting, diarrhoea and sun sensitivity.

Mode of action

1. **Bacteriostatic effect:** It stops bacteria from growing and multiplying rather than killing them outright.
2. **Gram-negative/positive action:** It is effective against both Gram-positive and Gram-negative bacteria.

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3. **Targeting ribosomes:** It enters the bacterial cell and binds reversibly to the 30S ribosomal subunit.
4. **Blocking protein production:** By binding to the 30S ribosome, it prevents aminoacyl-tRNA from binding to the acceptor site. This inhibits translation and protein synthesis.

Q.18 [Long] Describe fluoroquinolones and their mode of action.

Fluoroquinolones are synthetic antibacterial agents, derived as a by-product of chloroquine synthesis (chloroquine is a synthetic drug made by modifying quinine, a natural alkaloid of the bark of the South American *Cinchona* tree). They were discovered by George Lesher (1962). They are bactericidal antibiotics that inhibit bacterial DNA replication. Rare but significant side effects: tendon ruptures, peripheral neuropathy, drug resistance and cardiac issues.

Mode of action

1. **Bactericidal effect:** They trap the enzymes of DNA replication in a complex that causes double-strand DNA breaks. The bacteria cannot multiply and die.
2. **Targeting topoisomerases:** They enter the bacterial cell and inhibit its type II topoisomerases.
3. **Targeting topoisomerase IV:** mainly in Gram-positive bacteria; this stops the separation of replicated DNA.
4. **Targeting DNA gyrase:** mainly in Gram-negative bacteria; this stops the uncoiling of DNA during replication.

Q.19 [Short] Which enzymes are targeted by fluoroquinolones in Gram-positive and Gram-negative bacteria?

They inhibit the type II topoisomerases. **Topoisomerase IV** is the main target in Gram-positive bacteria (separation of replicated DNA stops). **DNA gyrase** is the main target in Gram-negative bacteria (uncoiling of DNA during replication stops).

Q.20 [Long] Describe sulfonamides and their mode of action.

Sulfonamides are synthetic compounds derived from sulfanilamide, discovered by Gerhard Domagk in 1932. They are bacteriostatic antimicrobial drugs that inhibit the bacterial synthesis of folic acid (folate), which is essential for DNA synthesis and bacterial growth. They are mainly used for urinary tract infections and skin infections and are often given orally. Side effects: rashes, nausea, photosensitivity and kidney damage.

Mode of action

1. **Bacteriostatic effect:** They do not kill bacteria immediately but inhibit their growth, so that the body's immune system can destroy them.
2. **Selective toxicity:** Bacteria must build folate from scratch, whereas humans simply absorb it from food. So the drug starves bacteria while leaving human cells untouched.
3. **Competitive inhibition:** They bind to the enzyme that makes folate, taking the place of para-amino benzoic acid (PABA) in the folate pathway.

4. **Combined therapy:** They are often combined with other agents, like trimethoprim, for a stronger bactericidal effect.
5. **Broad spectrum:** They are effective against both Gram-positive and Gram-negative bacteria (urinary tract infections, acne, respiratory infections).

The textbook sentence on competitive inhibition (p.212) is incomplete: "...essentially taking para amino benzoic acid pathway". The meaning is that the drug competes with para-amino benzoic acid (PABA) for the folate-making enzyme.

Q.21 [Short] Why are sulfonamides selectively toxic to bacteria?

Bacteria must build folate (folic acid) themselves, while humans simply absorb it from food. Sulfonamides block folate synthesis, so they starve the bacteria of this essential vitamin but leave human cells untouched.

Q.22 [Short] List the side effects of the four classes of antibiotics.

Antibiotic	Side effects
Penicillin	Nausea, vomiting, diarrhoea, abdominal pain, skin rashes
Tetracycline	Nausea, vomiting, diarrhoea, sun sensitivity
Fluoroquinolones	Tendon ruptures, peripheral neuropathy, drug resistance, cardiac issues (rare but significant)
Sulfonamides	Rashes, nausea, photosensitivity, kidney damage

23.3 ANTIVIRAL AND ANTIRETROVIRAL DRUGS

Q.23 [Short] Define antiviral and antiretroviral drugs.

Antiviral drugs	Antiretroviral drugs
Used to treat viral infections in general.	Designed to treat illnesses caused by retroviruses, including HIV.
Prevent the growth and multiplication of viruses within the body.	Stop HIV from proliferating and aid the immune system's defence.
Act against a wide range of viruses, e.g., hepatitis, influenza and herpes viruses.	Act against retroviruses (viruses with RNA as genetic material that replicate by reverse transcriptase).
All antivirals are not antiretrovirals.	All antiretrovirals are antivirals.

Q.24 [Short] What are retroviruses?

Retroviruses are viruses that have RNA as their genetic material and replicate by using the enzyme reverse transcriptase. HIV (Human Immunodeficiency Virus) is an example.

Q.25 [Short] "All antiretrovirals are antivirals but all antivirals are not antiretrovirals." Explain.

Retroviruses are one group of viruses, so a drug against them (antiretroviral) is also an antiviral. But many antivirals act on other viruses such as hepatitis, influenza and herpes viruses, which are not retroviruses; these are not antiretrovirals.

23.4 MONOCLONAL ANTIBODIES

Q.26 [Long] What are monoclonal antibodies? Describe their advantages over other drug classes.

Monoclonal antibodies are artificially laboratory-produced molecules that enhance the immune system's ability to fight against pathogens. They target specific antigens, which makes them effective for treating cancer, autoimmune disorders and infectious diseases. They are integral to personalized medicine, because treatment can be tailored to the profile of the individual patient.

Advantages

1. Compared with traditional drugs, they offer lower toxicity and the potential for long-lasting effects.
2. They have the least side effects compared with conventional chemotherapeutics.
3. They have high specificity, binding to a single target with minimal variability.
4. They provide targeted therapy in cancer, reducing damage to healthy tissues.
5. They are easy to prepare and are used for research purposes.
6. They reduce the risk of drug interactions.

Q.27 [Short] How are monoclonal antibodies related to personalized medicine?

They are integral to personalized medicine. They allow tailored treatment based on the profile of the individual patient, so the therapy is better suited to each patient's unique biology.

The textbook numbers the advantages as section 23.4.2, although there is no section 23.4.1.

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